

Coherent output clusters, frozen likelihood, and the adapted predictable-baseline boundary

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1 Purpose and scope

R-106 left the complete coherent output-frequency current as a legal coordinate for the remaining finite-cutoff, fixed-positive-floor $q = 10/9$ production problem. The present note determines the smallest coordinate on which that suggestion can be correct and audits the two equivalent Gibbs and direct-source formulations before another owner split is introduced.

The result has ten exact parts.

1. The endpoint likelihood tilt has a positive entropy-production identity and an exact root-martingale variance decomposition. These are identities, not a new free-energy bound.
2. A jointly frozen family of coherent-output Gaussian rows has an exact backward-resolvent density martingale. Its conditional entropy inequality is saturable and therefore has no spare reserve for revealing a same-root nonlinear coefficient.
3. One fresh-root packet with a past-measurable output map and mixed baseline closes by one $\det_2$ formula without output factorisation.
4. A bounded predictable future row has a strictly positive normalization defect, so rowwise predictability does not license the global frozen backward-resolvent formula. Stepwise conditional normalizers remain legal.
5. The coefficient trace has a canonical positive output allocation, but an exact one-Fourier-pair fixture proves that a single output frequency is not a signed atom. Cancellation occurs only across the contraction-connected cluster $\{0, 2, -2\}$.
6. Independent output determinant normalisers lose positive slack which grows linearly under repeated rank-one rows. Sequential Schur updates are mandatory.
7. The direct source action reduces exactly to the sum of complete predictable baselines plus one terminal sextic and one source-energy payment. Covariance mass against one deterministic predictable PSD weight is subdivision invariant.
8. An adapted sine fixture proves that separating the second-jet/forest companions creates two M^2 costs although their signed combination tends to zero.
9. A pure carrier-information bridge costs $-(d/2)\log t$ and is therefore unusable at the diagonal and across growing root dimension.
10. A useful Gaussian-divergence convexification identity is exact, but its natural inward-flow remainder has the unfavourable sign. That route is parked, not declared impossible.

The remaining theorem is one subdivision-invariant adapted complete-cluster or matrix-Carleson estimate after all visits, all input convolutions, the entire R-063 forest, the mixed baseline, and future feedback have been recombined. No such estimate is proved here. Hence $\text{OVERLAP}_{\text{src}}$, Nelson, removals, the interacting measure, and Sector A remain open. Tier stays T4.

2 Endpoint likelihood tilt and entropy production

Use the R-106 convention: $\nu_{J,0}$ is the self-coupled endpoint, $\nu_{J,1}$ is the controlled independent-carrier endpoint, and

$$\frac{d\nu_{J,0}}{d\nu_{J,1}} = \exp\{\mathcal{L}_J - (\Phi_{J,0} - \Phi_{J,1})\}. \quad (2.1)$$

To avoid confusing the new exponential parameter with the A9 interpolation time, define for $0 \leq s \leq 1$

$$\psi_J(s) := \log \mathbb{E}_{\nu_{J,1}} e^{s\mathcal{L}_J}, \quad \frac{d\rho_{J,s}}{d\nu_{J,1}} := e^{s\mathcal{L}_J - \psi_J(s)}. \quad (2.2)$$

Then $\rho_{J,0} = \nu_{J,1}$, $\rho_{J,1} = \nu_{J,0}$, and $\psi_J(1) = \Phi_{J,0} - \Phi_{J,1}$.

Theorem 2.1 (exact endpoint entropy production)

At every finite cutoff,

$$\psi_J'(s) = \mathbb{E}_{\rho_{J,s}} \mathcal{L}_J, \quad \psi_J''(s) = \text{Var}_{\rho_{J,s}}(\mathcal{L}_J) \geq 0. \quad (2.3)$$

Moreover

$$H(\nu_{J,0} \mid \nu_{J,1}) = \int_0^1 s \text{Var}_{\rho_{J,s}}(\mathcal{L}_J) ds, \quad (2.4)$$

$$H(\nu_{J,1} \mid \nu_{J,0}) = \int_0^1 (1-s) \text{Var}_{\rho_{J,s}}(\mathcal{L}_J) ds, \quad (2.5)$$

and hence

$$\Phi_{J,0} - \Phi_{J,1} = \mathbb{E}_{\nu_{J,0}} \mathcal{L}_J - \int_0^1 s \text{Var}_{\rho_{J,s}}(\mathcal{L}_J) ds, \quad (2.6)$$

$$\Phi_{J,0} - \Phi_{J,1} = \mathbb{E}_{\nu_{J,1}} \mathcal{L}_J + \int_0^1 (1-s) \text{Var}_{\rho_{J,s}}(\mathcal{L}_J) ds. \quad (2.7)$$

Proof. Differentiate (2.2). The two intermediate entropies are

$$H(\rho_{J,s} \mid \nu_{J,1}) = s\psi_J'(s) - \psi_J(s), \quad (2.8)$$

$$H(\rho_{J,s} \mid \nu_{J,0}) = (s-1)\psi_J'(s) - \psi_J(s) + \psi_J(1). \quad (2.9)$$

Their derivatives are respectively $s\psi_J''(s)$ and $(s-1)\psi_J''(s)$. Integrating from zero to one proves (2.4)–(2.7). $\square$

Let $(\mathcal{F}_j)$ be any finite root-reveal filtration exhausting the cutoff source, and define the $\rho_{J,s}$ martingale differences

$$d_j^{(s)} F := \mathbb{E}_{\rho_{J,s}}[F \mid \mathcal{F}_j] - \mathbb{E}_{\rho_{J,s}}[F \mid \mathcal{F}_{j-1}]. \quad (2.10)$$

Orthogonality gives

$$\boxed{\text{Var}_{\rho_{J,s}}(\mathcal{L}_J) = \sum_j \mathbb{E}_{\rho_{J,s}} |d_j^{(s)} \mathcal{L}_J|^2.} \quad (2.11)$$

Thus entropy production has a positive root decomposition. It does not bound the endpoint difference: the conditional laws in (2.10) depend on the unknown tilt, and (2.6)–(2.7) are exact rewritings of R-106's endpoint identity.

3 Jointly frozen coherent-output likelihood martingale

Let independent standard Gaussian roots be $\xi_1, \dots, \xi_N$, and let $A_r : E_r \rightarrow \mathcal{H}_{\text{out}}$ be jointly frozen maps into the complete coherent heat/output Hilbert space: conditionally on an initial sigma-field $\mathcal{F}_0$, every A_r is fixed and independent of all roots. Put

$$K := \sum_{r=1}^N A_r A_r^*, \quad K_{>j} := \sum_{r>j} A_r A_r^*, \quad R_{>j} := (\mathbf{I} + qK_{>j})^{-1}, \quad (3.1)$$

$$s_j := \sum_{r \leq j} A_r \xi_r. \quad (3.2)$$

The normalized Gaussian likelihood is

$$\mathcal{L}_A = \frac{1}{2} \log \det(\mathbf{I} + qK) - \frac{q}{2} \|s_N\|^2. \quad (3.3)$$

If

$$Q_A = \frac{1}{2} (\|s_N\|^2 - \text{Tr } K), \quad D_q(K) = \frac{1}{2} \{q \text{Tr } K - \log \det(\mathbf{I} + qK)\}, \quad (3.4)$$

then

$$\mathcal{L}_A = -qQ_A - D_q(K), \quad \mathbb{E}[e^{\mathcal{L}_A} \mid \mathcal{F}_0] = 1. \quad (3.5)$$

Theorem 3.1 (backward-resolvent density martingale)

Let $\mathcal{F}_j = \mathcal{F}_0 \vee \sigma(\xi_1, \dots, \xi_j)$. For $0 \leq j \leq N$,

$$\log M_j = \frac{1}{2} \log \det(\mathbf{I} + qK) - \frac{1}{2} \log \det(\mathbf{I} + qK_{>j}) - \frac{q}{2} \langle s_j, R_{>j} s_j \rangle. \quad (3.6)$$

Then

$$M_j = \mathbb{E}[e^{\mathcal{L}_A} \mid \mathcal{F}_j], \quad M_0 = 1, \quad M_N = e^{\mathcal{L}_A}. \quad (3.7)$$

Writing $m_j = \log(M_j/M_{j-1})$ gives

$$\mathbb{E}[e^{m_j} \mid \mathcal{F}_{j-1}] = 1. \quad (3.8)$$

For every conditional law λ_j of the fresh root with finite entropy relative to its standard Gaussian law γ_j ,

$$\mathbb{E}_{\lambda_j} m_j - H(\lambda_j \mid \gamma_j) \leq 0, \quad (3.9)$$

and equality is attained by the exact conditional exponential tilt.

Proof. Integrate the unrevealed Gaussian sum in (3.3). This gives (3.6). One more Gaussian integration, or the determinant lemma together with the Woodbury identity, proves (3.7)–(3.8). Equation (3.9) is the conditional Gibbs variational formula; its normalizing logarithm is zero by (3.8). $\square$

The saturation in (3.9) is load-bearing. Frozen determinant entropy cannot be spent again on the difference between a fixed A_r and the true same-root state-dependent production coefficient.

Proposition 3.2 (a predictable future row breaks the frozen normaliser)

The jointly frozen hypothesis cannot be replaced by the usual progressive condition $A_r \in \mathcal{F}_{r-1}$. Let ξ_1, ξ_2 be independent standard Gaussians and take the bounded scalar rows

$$A_1 = 1, \quad A_2 = c(\xi_1), \quad c(x) = \mathbf{1}_{\{|x|>1\}}. \quad (3.10)$$

Inserting the random total and tail covariances into the frozen candidate (3.6) gives, at the first reveal,

$$\widetilde{M}_1(x) = \left(\frac{1 + q(1 + c(x))}{1 + qc(x)} \right)^{1/2} \exp \left\{ -\frac{qx^2}{2(1 + qc(x))} \right\}. \quad (3.11)$$

After multiplication by the standard Gaussian density, the inner region has precision $\beta_0 = 1 + q = 19/9$ and the outer region has precision $\beta_1 = (1 + 2q)/(1 + q) = 29/19$. Hence

$$\mathbb{E}\widetilde{M}_1 = \operatorname{erf} \sqrt{\frac{19}{18}} + \operatorname{erfc} \sqrt{\frac{29}{38}} = 1 + \operatorname{erf} \sqrt{\frac{19}{18}} - \operatorname{erf} \sqrt{\frac{29}{38}} > 1, \quad (3.12)$$

because $19/18 > 29/38$ and erf is strictly increasing. The independent executable obtains

$$\mathbb{E}\widetilde{M}_1 = 1.070433115292664 \dots \quad (3.13)$$

from both the closed error-function formula and direct piecewise quadrature. The discontinuity is inessential. For the bounded smooth choice $A_2 = \epsilon \tanh \xi_1$, with $v = (1 + q)^{-1}$ and $\delta = \epsilon^2$, the same candidate satisfies

$$\frac{d}{d\delta} \mathbb{E}\widetilde{M}_1|_{\delta=0} = \frac{q^2}{2} \operatorname{Cov}_{N(0,v)}(X^2, \tanh^2 X) > 0. \quad (3.14)$$

Indeed $y \mapsto \tanh^2 \sqrt{y}$ is strictly increasing and $2 \operatorname{Cov}(Y, g(Y)) = \mathbb{E}[(Y - Y')(g(Y) - g(Y'))] > 0$. Thus the global backward-resolvent expression is not a density martingale for general smooth predictable future rows either. This is a method no-go, not a negative complete-action fixture. A genuinely progressive construction must multiply the actual one-fresh-root conditional normalisers rather than substitute random future rows into one frozen global determinant.

4 Coherent heat/output trace and root packet

Use R-082's heat lift

$$\Xi_\Sigma(z, y)(r) = C(z + r)y \in L^2(\mu_\Sigma; \mathbb{R}^4), \quad B = C^T C. \quad (4.1)$$

For endpoint H , root j , derivative direction i , and an independent root copy G_j of covariance $\Delta\Gamma_j$, define

$$\tau_{H,j,i,n} := \int \mathbb{E}_{G_j} \left| C_H(\widehat{\cdot, r}) \partial_i G_j(n) \right|^2 \mu_\Sigma(dr). \quad (4.2)$$

Proposition 4.1 (exact positive output allocation)

Parseval gives

$$\boxed{\sum_n \tau_{H,j,i,n} = \int_{\mathbb{T}^3} (P_\Sigma B_H) : \Delta\Gamma_{j,i} \, dx.} \quad (4.3)$$

In Fourier coordinates, if $Q_{j,i}(p)$ is the derivative-root covariance,

$$\tau_{H,j,i,n} = \int \sum_p \operatorname{Tr} \{ \widehat{C}_H(n - p, r) Q_{j,i}(p) \widehat{C}_H(n - p, r)^* \} \mu_\Sigma(dr). \quad (4.4)$$

The summands in (4.2) are nonnegative. They are not the conditional variances of the actual current when C_H depends on the same root. In that case R-082 (8.4) retains a signed posterior covariance defect.

Let

$$a_j = d_j \Xi_0, \quad f_j = d_j (\Xi_j - \Xi_0), \quad \iota_j = d_j (\Xi_* - \Xi_j). \quad (4.5)$$

Conditional expectation commutes with deterministic Fourier projection. Therefore R-079 (3.3)–(3.5) becomes the complete packet

$$\begin{aligned}\mathfrak{P}_j = & \sum_{i,n} \mathbb{E} \left[\operatorname{Re} \langle \hat{a}_{j,i,n}, \hat{f}_{j,i,n} \rangle + \frac{1}{2} |\hat{f}_{j,i,n}|^2 \right. \\ & \left. + \operatorname{Re} \langle \hat{a}_{j,i,n} + \hat{f}_{j,i,n}, \hat{\ell}_{j,i,n} \rangle + \frac{1}{2} |\hat{\ell}_{j,i,n}|^2 \right] \\ & - \frac{1}{2} \sum_{i,n} \mathbb{E} (\tau_{j,j,i,n} - \tau_{0,j,i,n}) \\ & - \frac{1}{2} \sum_{i,n} \mathbb{E} (\tau_{*,j,i,n} - \tau_{j,j,i,n}).\end{aligned}\tag{4.6}$$

All input convolutions reaching n are summed before the square. The mixed $f_{j-\ell_j}$ term is inside the second inner product. Expanding the two trace increments through R-063 (5.5) produces exactly the P_3/P_1 and $P_4/P_2^{\text{cross}}/\Sigma_j Q_j/P_0^{\text{cross}}/\mathcal{R}_{3,j}$ forest. The forest is an alternative realization of part of (4.6); adding it separately would double count it.

5 One-step predictable whole-output determinant with baseline

Let $A : E_j \rightarrow \mathcal{H}_{\text{out}}$ and $b \in \mathcal{H}_{\text{out}}$ be measurable before the one fresh root ξ , put $S = AA^*$, and define

$$L_{\text{fr}} = \frac{1}{2} \|A\xi + b\|^2 - \frac{1}{2} \operatorname{Tr} S.\tag{5.1}$$

Theorem 5.1 (whole-output det2 formula)

$$\boxed{\begin{aligned} \log \mathbb{E}[e^{-qL_{\text{fr}}} \mid \mathcal{F}_{\text{past}}] = & -\frac{1}{2} \log \det {}_2(\mathbf{I} + qS) \\ & -\frac{q}{2} \langle b, (\mathbf{I} + qS)^{-1} b \rangle \end{aligned}}\tag{5.2}$$

and

$$\log \mathbb{E}[e^{-qL_{\text{fr}}} \mid \mathcal{F}_{\text{past}}] \leq \frac{q^2}{4} \|S\|_{\mathfrak{S}_2}^2.\tag{5.3}$$

Proof. Complete the finite-dimensional Gaussian square. Then use $\det {}_2(\mathbf{I} + qS) = \det(\mathbf{I} + qS) e^{-q \operatorname{Tr} S}$ and $u - \log(1+u) \leq u^2/2$ for $u \geq 0$. $\square$

No output-frequency factorisation occurs in (5.2). If P_n is an output projection, S retains every off-diagonal block $P_n S P_m$. Thus one past-measurable fresh-root step closes, and jointly frozen rows may be iterated by Theorem 3.1. Proposition 3.2 forbids replacing the stepwise normalisers by one global frozen determinant when future rows depend on revealed roots. Cross-output same-root dependence remains in the nonfrozen defect.

6 A single output frequency is not a signed atom

Proposition 6.1 (exact one-pair cluster fixture)

On the normalized circle take

$$X(x) = a \cos x + b \sin x, \quad a, b \sim^{\text{iid}} N(0, \sigma^2), \quad z = \frac{a - ib}{2},\tag{6.1}$$

and the scalar linear-row current $J = X \partial_x X$. Then

$$\hat{J}(0) = 0, \quad \hat{J}(2) = iz^2, \quad \hat{J}(-2) = -i\bar{z}^2.\tag{6.2}$$

The independent-copy trace allocation is

$$\tau_0 = \sigma^2 |z|^2, \quad \tau_2 = \tau_{-2} = \frac{\sigma^2}{2} |z|^2. \quad (6.3)$$

With the physical outer factor $1/2$, the expected output packets are

$$n = 0 : -\frac{\sigma^4}{4}, \quad n = 2 : \frac{\sigma^4}{8}, \quad n = -2 : \frac{\sigma^4}{8}. \quad (6.4)$$

Only the complete cluster $\{0, 2, -2\}$ cancels:

$$\frac{1}{2} \sum_n (|\hat{J}(n)|^2 - \tau_n) = |z|^4 - \sigma^2 |z|^2, \quad (6.5)$$

$$\mathbb{E}[(6.5)] = 0, \quad \inf (6.5) = -\frac{\sigma^4}{4}. \quad (6.6)$$

Proof. Write $X = ze^{ix} + \bar{z}e^{-ix}$, convolve its two Fourier coefficients, and replace the differentiated factor by an independent copy for (6.3). Finally use $\mathbb{E}|z|^2 = \sigma^2/2$ and $\mathbb{E}|z|^4 = \sigma^4/2$. $\square$

The fixture embeds, up to a positive constant, in a linear Pauli–Fierz row. It refutes single-output positivity. It is not a negative complete-action or Nelson example.

7 Why independent output normalisers fail

For a positive semidefinite covariance T , let

$$D_q(T) = \frac{1}{2} \{q \operatorname{Tr} T - \log \det(\mathbf{I} + qT)\}. \quad (7.1)$$

For positive semidefinite blocks T_a ,

$$\log \det(\mathbf{I} + q \sum_a T_a) \leq \sum_a \log \det(\mathbf{I} + qT_a), \quad (7.2)$$

and therefore

$$D_q(\sum_a T_a) \geq \sum_a D_q(T_a). \quad (7.3)$$

The lost determinant slack is not harmless. For m copies of one rank-one block λP ,

$$S_m = \frac{1}{2} \{m \log(1 + q\lambda) - \log(1 + qm\lambda)\}, \quad (7.4)$$

$$\frac{S_m}{m} \longrightarrow \frac{1}{2} \log(1 + q\lambda) > 0. \quad (7.5)$$

This fires the scoped output-normalisation no-go: multiplying independently normalised row or output likelihoods can lose a linear amount. The exact replacement is the sequential Schur identity. With $K_m = \sum_{a \leq m} T_a$ and $R_{m-1} = (\mathbf{I} + qK_{m-1})^{-1}$,

$$\boxed{\log \det(\mathbf{I} + qK_m) - \log \det(\mathbf{I} + qK_{m-1}) = \log \det\{\mathbf{I} + qR_{m-1}^{1/2} T_m R_{m-1}^{1/2}\}.} \quad (7.6)$$

The increments depend on the reveal order, but their sum is the invariant whole-output determinant.

8 The irreducible same-root residual

Write the complete current, after the predictable part is frozen, as

$$Y(\xi) = b + A\xi + R(\xi). \quad (8.1)$$

Let $\delta\tau(\xi)$ be the difference between the complete heat-lifted trace and $\text{Tr}(AA^*)$.

Proposition 8.1 (exact frozen/nonfrozen split)

The complete covariance-normal packet is

$$L_{\text{full}} = L_{\text{fr}} + \mathfrak{R}_{\text{coh}}, \quad (8.2)$$

where

$$\boxed{\mathfrak{R}_{\text{coh}} = \text{Re}\langle b + A\xi, R(\xi) \rangle + \frac{1}{2}\|R(\xi)\|^2 - \frac{1}{2}\delta\tau(\xi).} \quad (8.3)$$

Proof. Expand $\|b + A\xi + R\|^2$ and subtract the frozen packet (5.1). $\square$

After the same-root Taylor/Wick conversion, $\delta\tau$ is precisely the complete R-063 forest. Future feedback belongs in R and its mixed pairing, not in an independently paid owner. Formula (8.3), coherently summed over a contraction-connected output cluster, is the minimal unresolved local object.

9 Direct predictable-baseline source-action normal form

On an R-104 temporally faithful chart, write

$$Z_k = S_\pi\xi + \sum_{\ell \leq k} S_\ell h_\ell, \quad Z_0 = S_\pi\xi, \quad (9.1)$$

and

$$\Delta_k V = V_J^{\text{ren}}(Z_k) - V_J^{\text{ren}}(Z_{k-1}), \quad P_k = \mathbb{E}[\Delta_k V \mid \mathcal{F}_{k-1}]. \quad (9.2)$$

R-077's complete backward-heat Doob reconstruction retains every fresh-root packet, including its R-063 forest, and gives it conditional expectation zero. Since $\mathbb{E}V_J^{\text{ren}}(Z_0) = 0$, the source action has the exact form

$$\boxed{\mathcal{A}_J(h) = \sum_k \mathbb{E}P_k + \frac{3}{20}\mathbb{E}\|Z_K\|_6^6 + \frac{9}{20}\mathbb{E}\sum_k \|h_k\|^2.} \quad (9.3)$$

Equation (9.3) is the direct counterpart of the Gibbs defect after the frozen martingale (3.6) is removed. It proves no lower bound on $\sum_k \mathbb{E}P_k$. It does show that a successful argument must retain the terminal sextic once, not once per visit, input leaf, or output frequency.

There is one exact subdivision-invariant predictable quantity. If $S_k S_k^* = \Delta C_k$, $\sum_k \Delta C_k = C_J$, and D is deterministic positive semidefinite, then

$$\boxed{\sum_k \|D^{1/2} S_k\|_{\text{HS}}^2 = \sum_k \text{Tr}(D \Delta C_k) = \text{Tr}(D C_J).} \quad (9.4)$$

This does not extend automatically to a same-root random D . The scalar fixture $D(G) = G^2$, $Q(G) = G^2 - 1$ gives

$$\mathbb{E}[D(G)Q(G)] = 3 - 1 = 2. \quad (9.5)$$

10 Termwise adapted second-jet estimates fail

Let $G \sim N(0, 1)$ and

$$h_M(G) = a \sin(MG), \quad r_M = e^{-2M^2}. \quad (10.1)$$

Direct Gaussian calculation gives

$$\mathbb{E}h_M^2 = \frac{a^2}{2}(1 - r_M), \quad (10.2)$$

$$\mathbb{E}(h'_M)^2 = \frac{a^2 M^2}{2}(1 + r_M), \quad \mathbb{E}(h_M h''_M) = -\frac{a^2 M^2}{2}(1 - r_M), \quad (10.3)$$

$$\mathbb{E}\{(h'_M)^2 + h_M h''_M\} = a^2 M^2 r_M, \quad (10.4)$$

and the Gaussian second-order Stein identity gives

$$\mathbb{E}[h_M^2(G)(G^2 - 1)] = \mathbb{E}[(h_M^2)'] = 2a^2 M^2 r_M. \quad (10.5)$$

Thus the source cost (10.2) stays bounded, while the two terms in (10.3) separately approach $\pm a^2 M^2/2$. Their signed sum (10.4) tends to zero. Termwise absolute or derivative-square control of the adapted forest is therefore impossible from the source cost alone. This is a method no-go; the complete signed second jet remains live.

11 Pure carrier information is not a diagonal bridge

For independent $x, y \sim N(0, I_d)$ let

$$z_t = \sqrt{1-t}x + \sqrt{t}y, \quad 0 < t \leq 1. \quad (11.1)$$

Then z_t is standard Gaussian and

$$\boxed{I(x; z_t) = -\frac{d}{2} \log t.} \quad (11.2)$$

Indeed, conditionally on x , z_t has covariance tI_d , and its covariance and mean contributions to the averaged Gaussian relative entropy are

$$\frac{d}{2}(t - 1 - \log t), \quad \frac{d}{2}(1 - t). \quad (11.3)$$

Their sum is (11.2). The cost diverges as $t \downarrow 0$. If a root block has dimension comparable to 2^{3j} , even a fixed $t < 1$ gives a non-summable dimensional cost. Hence pure carrier KL cannot identify the independent and same-root diagonal. This does not refute a coupled heat/covariance/forest interpolation.

12 A parked convexified Gaussian-divergence route

Use

$$\operatorname{div}_\gamma F = \operatorname{div} F - x \cdot F. \quad (12.1)$$

Suppose $\operatorname{div}_\gamma b = -2V$, let U be homogeneous of degree six, and set

$$F = b + \frac{1}{3} \nabla U. \quad (12.2)$$

Euler's identity $x \cdot \nabla U = 6U$ gives the exact convexification

$$\boxed{V + U = -\frac{1}{2} \operatorname{div}_\gamma F + \frac{1}{6} \Delta U.} \quad (12.3)$$

This identity is useful but does not have the desired exponential sign by itself.

To audit the most tempting flow shortcut, let $A = -\operatorname{div}_\gamma F$ and follow the inward flow $\dot{\Psi}_s = -F(\Psi_s)$. Under the usual finite-dimensional smooth diffeomorphism hypotheses, the change-of-variables remainder is

$$D_t = \int_0^t A(\Psi_s X) \, ds - tA(\Psi_t X). \quad (12.4)$$

Already for $F(x) = x$, $U = 0$, and $A = x^2 - 1$,

$$\Psi_s x = e^{-s} x, \quad D_t = c_t x^2, \quad c_t = \frac{1 - (1 + 2t)e^{-2t}}{2} > 0. \quad (12.5)$$

Indeed

$$\mathbb{E} e^{D_t} = \frac{1}{\sqrt{1 - 2c_t}} = \frac{e^t}{\sqrt{1 + 2t}} = \mathbb{E} e^{-t(X^2 - 1)}. \quad (12.6)$$

The natural contractive flow therefore produces a positive remainder which cannot be discarded for an upper bound. The negative $-(q/6)\Delta U(\Psi_t X)$ term may still help in the production problem, but no accepted theorem currently dominates (12.4) with it. We record the route as inconclusive and parked, not as a general flow no-go.

13 Minimal remaining adapted cluster theorem

Join two output frequencies when they share a root input, an R-063 contraction, or an off-diagonal covariance block. A legal output atom is a union of connected components of this graph. Proposition 6.1 proves that arbitrary singleton outputs are illegal; in the worst case the legal cluster is the whole root output.

Conditional target 13.1 (complete adapted cluster bound)

A sufficient new theorem would assign to each root its complete contraction-closed packet $\mathfrak{P}_j$, nonnegative allocations η_j, ζ_j , and deterministic loss c_j , such that

$$\mathbb{E} \left[\mathfrak{P}_j + \eta_j \mathcal{E}_j + \zeta_j \mathcal{Y}_{\text{parent}(j)} \right] \geq -c_j, \quad (13.1)$$

$$\sum_j \eta_j \leq \frac{9}{20}, \quad \sum_j \zeta_j \leq \frac{3}{20}, \quad \sum_j c_j < \infty. \quad (13.2)$$

The sextic in (13.1) is one unfragmented parent/root payment.

Equivalently, in the direct source-action form, one may prove constants $\eta < 9/20$, $\zeta < 3/20$, and $C < \infty$, uniform in cutoff, chart, and representation-preserving subdivision, for which

$$\sum_k \mathbb{E} P_k^{\text{comp}} \geq -\eta \mathbb{E} \sum_k \|h_k\|^2 - \zeta \mathbb{E} \|Z_K\|_6^6 - C. \quad (13.3)$$

Here P_k^{comp} contains the complete heat, covariance trace, mixed baseline, future feedback, rational current, all visits, and the full R-063 forest exactly once.

A determinant-compatible stronger form would prove for every contraction-closed cluster $\mathcal{C}$

$$\begin{aligned} \log \mathbb{E} \left[e^{-q(\mathfrak{P}_{j,\mathcal{C}} + \eta_j \mathcal{E}_j + \zeta_j \mathcal{Y}_{\text{parent}(j)})} \mid \mathcal{F}_{\text{past}} \right] \\ \leq \frac{q^2}{4} \|S_{j,\mathcal{C}}\|_{\mathfrak{S}_2}^2 + r_{j,\mathcal{C}}. \end{aligned} \quad (13.4)$$

with a summable matrix-Carleson ledger

$$\sum_{j,\mathcal{C}} \{\|S_{j,\mathcal{C}}\|_{\mathfrak{S}_2}^2 + r_{j,\mathcal{C}}\} < \infty. \quad (13.5)$$

Equations (13.1), (13.3), and (13.4) are targets, not theorems of this note. The operations must occur in this order: sum all visits, assemble every input convolution coherently at its output, restore the complete forest, and retain the terminal sextic once. Only then may one estimate.

14 Routes audited in this package

1. Endpoint entropy production: advanced as the exact formulas (2.4)–(2.11), but it returns the unknown endpoint free energy.
2. Jointly frozen multi-row likelihood: closed by (3.6); one past-measurable fresh-root determinant closes by (5.2). Their entropy inequalities are exactly saturable.
3. Global frozen normalisation for progressively predictable future rows: failed by the positive defect (3.12). Stepwise conditional normalisers remain legal.
4. Single-output positivity: failed by (6.4).
5. Independent output determinant normalisation: failed by the extensive slack (7.4)–(7.5); sequential Schur (7.6) survives.
6. Predictable-baseline direct action: advanced to the exact normal form (9.3); the complete baseline sum remains unbounded.
7. Termwise adapted forest: failed by (10.2)–(10.5).
8. Pure carrier KL identification: failed by (11.2).
9. Convexified Gaussian divergence: exact identity (12.3) advanced, but the naive inward-flow sign-dropping argument is parked by (12.5)–(12.6).

These verdicts refine the proof route. None is a counterexample to the complete production action or Nelson.

15 Devil’s-advocate audit

1. **Objection: the positive trace allocation makes each output packet nonnegative.**

UPHELD AGAINST THE CLAIM. Proposition 6.1 gives a strictly negative zero-output packet, compensated only by the two nonzero outputs.

2. **Objection: predictability alone licenses the global frozen backward-resolvent determinant, or one may condition on the same-root coefficient.**

UPHELD AGAINST THE CLAIM. Proposition 3.2 gives a bounded $\mathcal{F}_1$ -predictable future row for which the candidate density has mean greater than one; its smooth tanh companion has the same strict defect. Same-root conditioning also changes the root covariance: R-082 (8.4) leaves the signed posterior covariance defect, and R-093 shows that deterministic non-atomic coefficient revelation can have infinite information cost. Only jointly frozen multi-row formulas and one-fresh-root past-measurable conditional formulas are proved.

3. **Objection: the positive entropy production supplies an unused root budget.**

UPHELD AGAINST THE CLAIM. Equations (2.6)–(2.7) are the two exact orientations of the same endpoint difference. The frozen conditional entropy inequality (3.9) is saturable.

4. **Objection: the R-063 forest may be appended as an additional repair to (4.6).**

UPHELD AGAINST THE CLAIM. R-063 (5.5) is the exact Taylor/Wick realization of the trace/coefficient-curvature part already present in (4.6). Appending it duplicates ownership.

5. **Objection: independent output normalisers are conservative and therefore harmless.**

UPHELD AGAINST THE CLAIM. Their lost slack is positive and grows linearly by (7.5), precisely at the scale where summability is needed.

6. **Objection: derivative-square bounds can estimate the adapted forest term by term.**

UPHELD AGAINST THE CLAIM. The two companions in (10.3) have opposite M^2 growth under bounded source cost. Only the signed complete second jet cancels.

7. **Objection: R-063’s continuum bounds already imply (13.5).**

UPHELD AS UNSUPPORTED. The accepted weighted Sobolev convergence does not provide the unweighted, adapted, contraction-closed output matrix-Carleson sum. Available absolute dyadic majorants do not establish it.

8. **Objection: the convexified divergence identity proves a monotone transport bound.**

UPHELD AS UNSUPPORTED. The exact linear fixture (12.5) gives the natural inward-flow remainder the opposite sign from the desired discard. No production-specific domination of that remainder has been proved.

16 Authorities and scope firewall

The load-bearing authorities are R-063 (balanced Taylor/Wick forest), R-077 (complete backward-heat Doob packet), R-079 (exact root packet and ownership), R-082 (Pauli–Fierz square, heat lift, and covariance criterion), R-093 (source-union/Gibbs boundary), R-097 and R-100 (terminal covariance debt), R-104 (fixed-chart source-action assembly), R-105 (frozen determinant and subdivision boundary), and R-106 (endpoint likelihood and coherent-output successor).

Proved here are the exact finite-cutoff entropy-production identities, the jointly frozen backward-resolvent likelihood martingale, the one-fresh-root past-measurable determinant with mixed baseline, and the progressive-future-row normalisation no-go; coherent trace allocation, the one-pair output-cluster fixture, sequential determinant boundary, predictable-baseline action and covariance-mass identities, adapted second-jet and pure carrier-KL method no-gos, the convexified divergence identity, and the exact same-root residual.

Not proved here are (13.1), (13.3), (13.4), or any equivalent production adapted complete-cluster estimate; $\text{OVERLAP}_{\text{src}}$; the $q = 10/9$ Nelson estimate; cutoff or floor removal; an interacting measure; Sector-A closure; or a T5–T7 promotion. The next proof must establish one subdivision-invariant signed complete-cluster/matrix-Carleson inequality, not create another visit, leaf, output, forest, entropy, or sextic owner.

17 Executable verification

The primary certificate uses exact symbolic Gaussian, Fourier, determinant, covariance, second-jet, information, and divergence algebra. The independent certificate imports no symbolic, array, scientific, or

primary module: it uses exact fractions, a finite-tree Doob reconstruction, hand 2×2 determinants, radial Gaussian moments, and direct Gaussian quadrature.

Run

```
E:\Dev\TECT.venv\Scripts\python.exe '
codes/foundations/a13_classii_coherent_output_cluster_predictable_baseline_boundary_verify.py
```

from the repository root. The integrated verifier reruns both certificates and checks the note, PDF, manifest, authority pins, public surfaces, exploration ledger, negative registry, and no-overclaim firewall.

Result footer

Result ID: A13-CLASSII-COHERENT-OUTPUT-CLUSTER-PREDICTABLE-BASELINE-BOUNDARY

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

Ledger ID: R-107

Status: T4 ANALYTIC/EXACT/EXECUTED BOUNDARY; NELSON AND SECTOR A OPEN

Scope: Finite cutoff, fixed positive floor, $q=10/9$; jointly frozen
multi-row or one-step past-measurable coherent-output Gaussian
rows and exact scoped method fixtures.

Precise statement: exact likelihood-tilt entropy production, frozen
backward-resolvent martingale, progressive-row normalisation
boundary, and one-step whole-output det2 bound,
coherent trace allocation, one-pair output-cluster theorem,
sequential determinant boundary, predictable-baseline action,
covariance mass, second-jet and carrier-KL method no-gos, and
convexified Gaussian-divergence identity.

Dependencies: A9; R-063, R-077, R-079, R-082, R-093, R-097, R-100,
R-104, R-105, R-106.

Evidence grade: ANALYTIC/EXACT/EXECUTED; TWO NON-IMPORTING ROUTES.

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe '
codes/foundations/a13_classii_coherent_output_cluster_predictable_baseline_boundary_verify.py
```

Expected output:

Primary 80/80; independent 139/139; integrated 239/239;
aggregate 458/458 PASS.

Negative results: NG-2026-07-28-A13-PREDICTABLE-MULTIROW-BACKWARD-RESOLVENT;
NG-2026-07-28-A13-SINGLE-OUTPUT-FREQUENCY-PACKET;
NG-2026-07-28-A13-INDEPENDENT-OUTPUT-DETERMINANT-NORMALIZATION;
NG-2026-07-28-A13-ADAPTED-SECOND-JET-TERMSEPARATION;
NG-2026-07-28-A13-PURE-CARRIER-KL-DIAGONAL-BRIDGE.

Falsification gate: failure of any entropy, martingale, determinant, trace,
cluster, baseline, covariance, second-jet, information,
divergence, evidence, hash, PDF, ledger, or scope assertion.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement:

the adapted complete-cluster/matrix-Carleson inequality is open;
no OVERLAP_src, Nelson, removal, interacting-measure, Sector-A,
or T5--T7 conclusion follows.

Next required action: prove one subdivision-invariant adapted complete
contraction-closed cluster estimate with all owners retained once.